

Altered rectal perception is a biological marker of patients with irritable bowel syndrome.



BACKGROUND & AIMS: Lowered visceral perception thresholds have been suggested as a biological marker of irritable bowel syndrome (IBS). The current study sought to determine the prevalence of altered rectal visceral perception in patients with IBS and the correlation of altered perception thresholds with subjective symptoms. METHODS: Anorectal manometry and rectal perception thresholds to balloon distention were determined in 100 patients with IBS and 15 control subjects. Gastrointestinal and psychological symptoms were assessed by questionnaire. Perception thresholds and symptoms were reassessed after 3 months in 15 patients with IBS. RESULTS: Ninety-four percent of patients showed altered rectal perception in the form of lowered thresholds for aversive sensations (discomfort), increased intensity of sensations, or altered viscerosomatic referral. Hypersensitivity was found only for aversive sensations in response to rapid phasic distention; stool thresholds and thresholds in response to slow ramp distention were normal. Cluster analysis by physiological parameters identified three IBS subgroups with predominant patterns of symptoms. Longitudinal evaluation indicated a correlation between changes in perception thresholds and symptom severity. CONCLUSIONS: Because altered rectal perception is present in almost all patients with IBS and perception thresholds correlate with temporal changes in retrospective symptom severity, altered rectal perception represents a reliable biological marker of IBS.